NATIONAL INSTITUTE OF TECHNOLOGY PATNA, PATNA

(Ashok Rajpath Patna-800 005)

B. Tech. End-Sem Examination May-2025

Program: B.Tech (CSE&DD) *Department: APME*

Time allotted: ***3hrs.*** *Full Marks:* ***60***

SUBJECT: ***Engineering Physics*** *Course code:* ***PH24101***

All questions carry equal marks

1. (a) Discuss the theory of damped oscillation and explain its three conditions [9+3]
   (b) For a damped oscillation m=500 gm, k=100 N/m, and the damping coefficient b=75 gm/sec. What is the ratio of amplitude of damped oscillation to the initial amplitude at the end of 20 cycles?

2. (a) What is optical resonator? Describe the construction and working of Ruby laser with a neat energy level diagram. [9+3]
   (b) A transition between the energy levels between E_2 and E_1 produces a light of wavelength 632.8 nm, calculate the energy of the emitted photons.

3. (a) Establish the conservation of electromagnetic energy theorem in a region. [9+3]
   (b) A point source emits light with power 1000 Watt. Find the average value of Poynting vector at a distance of 2m from the source.

4. (a) What is the difference between Fraunhofer and Fresnel class of Diffraction? [9+3] (any three points). Describe an expression of intensity at a point for Fraunhofer diffraction due to a single slit. Draw the intensity distribution curve for maxima and minima.
   (b) In a single slit diffraction experiment, the first minimum for $\lambda_1 = 660$ nm coincides with the first maxima for wavelength λ_2 Calculate λ_2.

5. (a) Explain the de-Broglie hypothesis of matter wave. Assume the wavelength measurement of a photon gives $\lambda = 600$ nm to an accuracy to one part in a million, determine the minimum uncertainty in the position of the photon. [7+5]
   (b) Setup time independent Schoredinger wave equation for matter wave.
